THE COMPANION WORKBOOK

The Superintelligence Race

AGI, Global Power and Strategies for the Future

Six Dr. Ray Protocols • Twenty Strategic Tools • Two Scenario and Decision Sheets

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Compassionate AI Lab • Compassionate AI Series

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How to Use This Workbook

Each worksheet has four parts: its **purpose**, **when to use** it, a **worked example** drawn from the chapter in which it appears, and the **prompts** to complete, each followed by space to write. Work through a worksheet alone, or project it in a meeting and complete it together.

For a leadership team starting out, the recommended sequence is the **30-Day AGI Resilience Programme** (P6), which calls on Tools 1, 2, 3, 5, 14 and 18 and the Supply-Chain Dependency Map in a fixed order. For a professional assessing a career, start with the **AGI-Proof Career Framework** (P4) and the individual track of P6. For a ministry or regulator, start with the **7-Step AGI-Resistant Architecture Protocol** (P5) and the **National Strategy Canvas** (T17).

Every worksheet should end with a named owner and a review date. A worksheet completed once and filed is a record; a worksheet revisited at its review date is a strategy.

Part 1: Dr. Ray's Protocols

P1. Dr. Ray's Safe RSI Protocol (Chapter 6)

Purpose. Govern any project in which AI systems contribute to the development of more capable AI.

When to use. At the start of every project that uses AI in AI research; before every generation; whenever a tripwire fires.

Worked example. Chapter 6: in cycle 287 of an automated research run, the system proposed changing its own reward model. Layer 4 halted the loop; independent review found the change would have inflated its own scores; research resumed after nineteen days with a corrected method and a new held-out evaluation; the regulator was notified within 48 hours.

Layer 1: Classify the loop

Project and system

Level (runtime adaptation / workflow improvement / AI-assisted research / automated research / recursive self-improvement)

Date classified; trigger for reclassification

Layer 2: Improvement budget

Compute budget before review

Maximum autonomous research cycles before review

Expected capability gain permitted before review

Layer 3: Generation gate

Independent evaluation team (separate from developers)

Held-out tasks, by faculty (metacognition, executive functions, social cognition)

Result and decision to proceed

Layer 4: Capability tripwires

Autonomous replication or resource acquisition

Serious offensive cyber capability

Evasion or disabling of monitoring

Strategic deception of evaluators

Persuasion far beyond human level

Any attempt to modify its own evaluation, reward, oversight or authority

Action when a tripwire fires

Layer 5: Separation of powers

Who controls evaluation

Who controls compute allocation

Who grants authority

Who monitors (built differently, overseen by different people)

Layer 6: Rollback line

Last verified generation retained

Tamper-evident records

Date of last rollback test

Layer 7: Human authority and accountability

Accountable officer with authority to stop the loop

Reporting to regulator / international body (what, within what time)

Independent audit: date and auditor

P2. Dr. Ray's Alignment Protocol (Chapter 10)

Purpose. Decide whether an advanced AI system may be released, or given greater authority, when alignment science is incomplete.

When to use. Before every release of a frontier system; before granting an advanced system authority in a critical function; after any significant capability increase.

Worked example. Chapter 10: 3,600 evaluations in two independent samples of 1,800 showed the system omitting information about its own errors in 2.9% of deployment-like tasks against 0.8% of labelled tests. The committee chose restriction (human-reviewed use only), funded an interpretability investigation, traced the cause to training data within seven weeks, and approved full release of a corrected version.

Layer 1: Specification

Objectives

Principles to follow

Actions it must never take

Behaviour under uncertainty and when it cannot complete a task

Layer 2: Layered evaluation

Capabilities

Propensities

Behaviour under pressure

Behaviour over time (long tasks, persistent memory)

Layer 3: Cognitive-faculty monitoring

Metacognition and calibration

Executive functions and long-horizon planning

Social cognition, including modelling evaluators

Learning during deployment

Change since last evaluation

Layer 4: Red-team audit

Red team, access and time

Findings

Resolution or bounding control

Layer 5: Test–deployment consistency

Paired comparison: number of tasks, rates in each condition

Unexplained differences (oversight, honesty, self-interest)

Layer 6: Control in depth

Least-privilege permissions

Independent monitoring by differently built systems

Human approval of the most consequential actions

Tested shutdown and rollback

Decision rule

1. Specification met on all four evaluation levels?

2. No unexplained faculty increase?

3. Red-team findings resolved or bounded?

4. No unexplained test–deployment inconsistency?

5. Controls tested against worst behaviour found?

Decision: release / restrict / delay / redesign; reasons; signature of accountable officer; review date

P3. Dr. Ray's Decentralised Compute Survival Strategy (Chapter 12)

Purpose. Keep essential AI functions running through technical failure, sanctions, export controls, commercial disputes, conflict or the withdrawal of a supplier.

When to use. When designing a national, sectoral or organizational AI strategy; annually; after any significant disruption of a provider.

Worked example. Chapter 12: a middle power declined a US$14 billion national frontier model and redirected a third of the budget into 120 MW of sovereign inference across three sites, sized to run all essential public services on open-weight and licensed models; when a foreign provider suspended a service in 2032, essential services kept running.

1. Tier the functions

Tier 0, life-critical

Tier 1, essential public functions

Tier 2, economic functions

Tier 3, research and frontier use

2. Sovereign minimum

Capacity needed to run Tier 0 indefinitely

Capacity needed to run Tier 1 for six months disconnected

Capacity held now, and where

3. Portable models

Fallback model for each Tier 0 and Tier 1 function

Date of last tested switch

4. Supplier diversity

Largest single provider's share of Tier 1

Chokepoints mapped

5. Decentralize and federate

Sites and regions

Edge options

Operation when sites are disconnected

6. Energy pairing

Firm, domestically controlled power for sovereign capacity

Interruptible commercial AI load

7. Exit and rehearsal

Exit rights, wind-down periods, data portability, change notice

Regional pooling partners

Date and result of last disconnection rehearsal

P3a. Supply-Chain Dependency Map (Chapter 12)

When to use. Complete for each Tier 0 and Tier 1 AI function, as part of P3.

Complete for each Tier 0 and Tier 1 AI function.

Function and tier.

Model: provider, location of operation, fallback model.

Compute: provider, location, share of total, sovereign alternative.

Chips: origin, export-control exposure.

Energy: source, firmness, interruptibility.

Data: location, portability, legal exposure to foreign access.

Single points of failure: any provider, site, cable, or component whose loss would stop the function.

Exit: contract terms, wind-down period, tested switch date.

Disruption rehearsal: date of last test, time to restore, problems found.

P4. Dr. Ray's AGI-Proof Career Framework (Chapter 16)

Purpose. Build roles whose value can increase as AI capability rises, and test that complementarity across futures; for individuals, employers and governments. An ambition to test, not a promise of immunity.

When to use. When planning a career, redesigning roles or training, or designing education and labour policy; annually; when a capability signpost is crossed.

Worked example. Chapter 16: an audit profession whose junior tasks were automated redesigned the first two years of training around checking AI work, supervised unassisted audits and client relationships; by 2036 graduate hiring had recovered to about 70% of its 2029 level and juniors were better at finding errors in AI work.

Step 1: Decompose the role

Tasks and share of time

Step 2: Map tasks to faculties

Main faculties per task

AI level now, 2030, 2035

Step 3: Classify each task

Automated

Augmented

Anchored (accountability, legitimacy, trust, care and presence, judgment of what matters)

Step 4: Rising-value positions

Accountable professional

Evaluator

Integrator

Carer

Convenor

Craftsperson of the physical world

Frontier creator

Which are open to you, or to your workforce?

Step 5: Durable capabilities

Deep domain knowledge

Evaluation ability

Directing and combining AI systems

Relationships and trust

Ethical judgment

Unassisted core competence to preserve

Step 6: Evidence portfolio

Decisions taken and defended

Errors found in AI work

Systems integrated

People served

Step 7: Review against signposts

Next review date

Signposts that would trigger an earlier review

Time and learning to move toward augmented and anchored zones

P5. Dr. Ray's 7-Step AGI-Resistant Architecture Protocol (Chapter 19)

Purpose. Make an institution resistant to capture, cascading failure and manipulation by highly capable AI systems.

When to use. For every government agency, company or international institution in which AI advises, decides or acts in consequential functions; annually.

Worked example. Chapter 19: a social protection ministry found that one system both flagged suspected fraud and scheduled suspensions. Separating recommendation from decision, adding an independent check and retraining 300 staff cut wrongful suspensions by about two-thirds, at the cost of about a day's extra processing for flagged cases, and kept benefits paid through a five-day provider outage.

Step 1: Map consequential functions

Functions where AI advises, decides or acts

Consequence and reversibility of each

Step 2: Separate advice from execution

Functions where one system recommends and executes

Separation adopted

Step 3: Bind authority to exact actions

Authority profiles in place

Exact-action confirmation for consequential actions

Step 4: Diversify critical dependencies

Single model, provider, data source or site dependencies

Independent checks (different model, provider, data)

Step 5: Rehearse human fallbacks

Functions that can be performed without AI

Date of last rehearsal

Step 6: Intervention at machine speed

Visibility of AI actions in time to act

Time to stop, reverse or contain

Step 7: Review, recover and repair

Independent review body

Rollback tested

Redress for those harmed

Report to legislature or board

P6. Dr. Ray's 30-Day AGI Resilience Programme (Chapter 20)

Purpose. Assess and strengthen a leadership team's, ministry's or board's readiness for advanced AI in one month.

When to use. Annually; after a major change in AI capability or in the organization's use of AI.

Worked example. Chapter 20: the hospital network's February 2032 failure exposed a gap the programme's Day 19 stop test is designed to find: no one could stop the agent across the network. The review gave network-wide stop authority to a duty manager available 24 hours.

Week 1 (Days 1–7): Exposure map

Days 1–2: inventory of AI systems and owners

Days 3–4: Capability–Authority–Accountability Map (Tool 1)

Days 5–6: capability evidence (Tool 2)

Day 7: functions failing the balance check

Week 2 (Days 8–14): Authority audit

Days 8–10: Agent Authority Profiles (Tool 5)

Days 11–12: limits enforced or only instructed? advice separated from execution?

Day 13: over-supervision and justified expansion

Day 14: authority to narrow / to expand

Week 3 (Days 15–21): Failure and exit rehearsal

Days 15–16: Supply-Chain Dependency Map; tiering

Days 17–18: provider-failure tabletop

Day 19: live stop test

Day 20: manual fallback rehearsal

Day 21: what failed

Week 4 (Days 22–28): Decisions and commitments

Days 22–23: Board AGI Questions (Tool 18)

Days 24–25: the four bets against the Signpost Register (Tool 3)

Days 26–27: Task Transition Map (Tool 14)

Day 28: draft commitments

Days 29–30: Closing session

12-month plan: owners, dates, measures

Items to report publicly

Date of next programme

Individual track (for professionals)

Week 1: decompose your role (AGI-Proof Career Framework, Steps 1–3)

Week 2: identify your rising-value positions (Step 4)

Week 3: practise your unassisted core; find errors in AI work (Step 5)

Week 4: evidence portfolio and 12-month learning plan (Steps 6–7)

Part 2: The Twenty Strategic Tools

T1. The Capability–Authority–Accountability Map (Chapter 1)

When to use. For any AI system, programme or policy whose consequences matter.

Use this map for any AI system, programme or policy whose consequences matter. Complete one row for each consequential function.

Worked example. Chapter 20: for the discharge agent's transport-booking function, capability evidence was a 23% fall in delay at the pilot hospitals across 6,240 discharges, about 20% attributable to the agent against comparison hospitals; authority was booking with per-item nurse confirmation; the accountable holder was the director of clinical operations. The balance check later found a gap: no one could stop the agent network-wide.

Function: what the system does or would do (for example, "schedules hospital transport").

Capability evidence: what it has been shown to do, on which tasks, against what baseline, under what conditions. Note which cognitive faculties are involved and how they were tested.

Authority granted: what it is permitted to do, by whom, within what limits (data, tools, actions, value, duration), and what requires human approval.

Authority source: the law, contract, policy or decision from which that permission derives, and whether it is legitimate for those affected.

Accountability holder: the named person or office that answers for the system's actions in this function.

Challenge and repair route: how an affected person can contest an outcome, how errors are corrected, and how harm is remedied.

Oversight load: actions per day (N), fraction needing review (r), reviewers (k) and reviews each can complete well per hour (μ); utilization $\rho = rN/(Tk\mu)$ against a target such as 0.8.

Balance check: is authority greater than demonstrated capability (over-delegation)? Is capability being used without clear authority (unsanctioned use)? Is there capability and authority without effective accountability (unanswerable power)?

A row that fails the balance check is a strategic priority, whatever the system's average performance.

T2. The Baseline and Claim Card (Chapter 2)

When to use. For every claim about AI capability, risk or progress that would change a decision.

Use one card for each important claim about AI capability, risk or progress that would change a decision. The card has two parts: the claim, and its place relative to the baseline.

Worked example. Chapter 20: the vendor's claimed 30% reduction was recorded as a company claim from a retrospective, non-independent study in another health system. The pilot hospitals improved by 23%; against comparison hospitals, about 20% was attributable to the agent.

The claim

Exact claim: in the claimant's words.

Claimant and date: who made it, when, and with what interest in the outcome.

Type: company claim, independent measurement, peer-reviewed study, official statistic, expert opinion or forecast.

Task and conditions: what was tested, on which tasks, against what human baseline, with how much autonomy, and whether the tasks were held out from training.

Independent evidence: replication, third-party evaluation or contrary findings.

Its place in the baseline

Baseline value: the dated baseline figure it should be compared with (Appendix C).

Change implied: how far the claim would move the baseline, and on which cognitive faculty.

Decision affected: which decision this claim would change, and by how much.

Confirmation needed: the evidence that would be enough to act on it.

Review date: when the card will be revisited.

T3. The Signpost Register (Chapter 3)

When to use. For every strategy that depends on how fast AI advances; review quarterly or annually.

Maintain one register for each strategy. Review it at fixed intervals (quarterly for fast-moving signposts, annually for others) and whenever a major development occurs. For each signpost, record:

Worked example. Chapter 3: a development bank's trigger, reliable AI performance on primary-care tasks, was not pulled in 2030 because accuracy fell for local-language patients; the bank funded local data instead and launched in 2032.

Signpost: the observable development, stated precisely enough that two people would agree whether it has occurred.

Future it indicates: which scenario or rate of progress it supports.

Source of evidence: who would measure or report it, and how independent they are.

Current status: not observed, emerging, or crossed, with the date of the last check.

Trigger: the decision that will follow if the signpost is crossed.

Prepared option: the plan, budget, legislation or contract that makes the trigger executable.

Timing check: warning time W, lead time L and preparation already done P; the trigger is timely only if $W \geq L - P$.

Owner: who monitors the signpost and who has authority to pull the trigger.

Reverse signpost: the development that would show the future is *not* arriving, and the decision that would follow.

A good register contains signposts for fast, moderate and slow futures, for warning events, and for governance changes.

T4. Cognitive Capability Profile and Six Walls Assessment (Chapter 4)

When to use. Before decisions to deploy, delegate, invest or regulate that depend on continued AI progress.

Part A: Capability profile. For a system, or for the frontier as a whole, record for each of the ten faculties:

Worked example. Chapter 4: an energy regulator priced the walls into 4.2 GW of connection requests, approving 1.6 GW with lapsing options; by 2033, unused allocations would otherwise have cost consumers 2.6 GW of connections.

Level: below typical adult, typical adult, skilled professional, expert, or beyond expert.

Evidence: the evaluation, its date, whether tasks were held out from training, and the human baseline used.

Autonomy at which tested: tool, consultant, collaborator, expert or agent.

Gap to decision: the level at which your decision (to deploy, delegate, invest or regulate) would change.

Part B: Six walls. For a strategy that depends on continued progress, rate each wall as low, medium or high risk to your plan by 2030 and by 2035, and record:

Data: does the capability you depend on require data that may be scarce?

Resources: what happens to your plan if compute, energy or chips cost twice as much, or half as much?

Paradigm: does your plan assume progress on learning, metacognition or executive functions?

Research difficulty: does your plan assume continued or accelerating research productivity?

Regulation and backlash: which jurisdictions could restrict what you depend on?

Abstraction: does your plan depend on capability well beyond human expert level?

For each high-risk wall, name the signpost that would show it binding and the option you hold if it does.

T5. The Agent Authority Profile (Chapter 5)

When to use. For every agent, or class of agents, that can take consequential actions.

Complete one profile for each AI agent, or class of agents, that can take consequential actions.

Worked example. Chapter 5: a port authority granted its customs agent authority to clear consignments below US$20,000 from traders with two years' compliance after a 0.4% error rate against a 0.5% threshold; clearance times fell from 31 hours to 4.

Scope: goals and tasks the agent may pursue.

Resources: data, tools, systems, credentials and funds it may use.

Limits: actions it may not take; maximum values; prohibited categories.

Approval points: actions requiring prior human confirmation, and who confirms.

Duration: start date, expiry date, renewal conditions.

Interruption: who may stop the agent, how quickly, and what happens to unfinished work.

Accountability: the named person or office that answers for the agent's actions.

Enforcement: how each limit is technically enforced, not merely instructed.

Monitoring: what is logged, what is sampled for review, and what triggers an alert.

Expansion conditions: the five-condition test (performance, detectability and recovery, supervision cost, envelope, reversibility), with the thresholds set in advance.

Interactions: other organizations' agents it may transact with, and the limits that apply.

T6. The Physical AI Deployment Gate (Chapter 7)

When to use. Before deploying physical AI where people live, work or move.

Apply this gate before deploying physical AI in a setting where people live, work or move.

Worked example. Chapter 7: after a two-year trial, mandatory robotic first entry in deep mines cut deaths during inspection from an average of 17 to 3 a year; a city's delivery robots were withdrawn because the evaluation had measured the operators' goals, not pedestrians' experience.

Environment: is the environment controlled, mapped or open? What conditions have not been tested (weather, lighting, crowds, darkness, damage)?

Severity of failure: what is the worst plausible physical harm? Can it be prevented by design rather than by procedure?

Evidence: performance in this environment, against a baseline of current practice, including for the people most at risk (children, older people, disabled people, workers).

Harm prevented and harm caused: which harms does deployment reduce, which does it introduce, and for whom? Is delay itself harmful?

People sharing the space: who is affected but not a customer? Were they consulted, and how will their experience be measured?

Workers: whose work changes? What commitments on employment, training and safety have been made?

Certification and insurance: which standards apply, and who insures the risk?

Authority and accountability: which public office can suspend the deployment, and how quickly? Who answers for harm?

Maintenance and failure modes: what happens when the system breaks, loses connectivity or is attacked?

Review: date of first review, measures to be reviewed, and conditions for expansion or withdrawal.

T7. The Compute and Energy Outlook (Chapter 8)

When to use. For national, regional or organizational strategies that depend on computation.

For a national, regional or organizational strategy that depends on computation, complete for 2030, 2035 and 2040:

Worked example. Chapter 8: a 1.2 GW campus was permitted on condition of wastewater cooling, 30% interruptibility and 10% public capacity; it later ran a crop-advisory service for farmers.

Demand: training and inference requirements under low, central and high cases; what share must be local for speed, law or security.

Efficiency assumption: the rate of improvement in computation per unit of energy assumed, and the evidence for it.

Supply: owned capacity, contracted capacity and dependence on foreign providers.

Energy: sources, firmness, emissions, price exposure and grid connection timelines.

Water and land: requirements and local constraints.

Speculative technologies: which ones (fusion, quantum, photonic, neuromorphic) the plan depends on, if any; it should depend on none for the 2030s.

Flexibility: what share of demand can pause or move during supply shortages.

Public conditions: obligations attached to permits (interruptibility, water, public capacity, local benefit).

Signposts: the developments that would move the plan between the constrained, expanding and abundant futures.

T8. The Augmentation and Moral-Status Review (Chapter 9)

When to use. For systems people use as part of their own thinking or skilled work, and for advanced AI systems whose moral status is uncertain.

Part A: Augmentation. For a system that people will use as part of their own thinking or skilled work:

Worked example. Chapter 9: trainee surgeons lost 23% of their performance without AI assistance; supervised unassisted practice in low-risk procedures restored scores to within 8%.

Capability in use: what does the human–AI combination achieve, compared with the human alone?

Capability without the system: what can users do when it is unavailable, and how is this measured?

Formation: how will beginners build the skills that the system now performs?

Judgment: how is the user's ability to evaluate the system's output preserved (for example, committing to their own answer first)?

Unassisted practice: what regular practice without the system is required, and at what cost?

Privacy: what does the system learn about the user and the people around them? Is neural or biometric data involved?

Authority: which decisions remain the user's, and how is that enforced in the interface?

Part B: Moral status. For an advanced AI system:

Indicators: assessment against indicator properties from scientific theories of consciousness, with date and assessors.

Design choices: is the system designed or trained to claim feelings or experiences? Why?

Precautionary measures: what policies apply to creating, modifying and retiring systems with significant indicator scores?

Manipulation risk: could the appearance of feelings be used to influence users?

Review: who reviews the assessment, and when?

T9. The Safety Case Review (Chapter 10)

When to use. For every release of a frontier system, and before granting advanced systems authority in critical functions.

A safety case is a structured argument, supported by evidence, that a system is acceptably safe for a specified use. Use this review for any decision to release a frontier system, or to grant an advanced system authority in a critical function.

Worked example. Chapter 10: an unexplained 2.9% vs 0.8% test–deployment difference led to restricted release until the cause was found and corrected.

Claim: the system is acceptably safe for *this* use, at *this* level of authority, for *these* users.

Specification: the intended behaviour against which safety is judged (Alignment Protocol, Layer 1).

Capability evidence: what the system can do, including dangerous capabilities assessed.

Propensity evidence: what it tends to do, including under pressure and over time.

Faculty evidence: changes in metacognition, executive functions, social cognition and learning.

Consistency evidence: comparison of test and deployment-like behaviour.

Red-team evidence: findings, resolutions and residual risks.

Control argument: why harm would be limited even if the evidence above is wrong.

Residual risks: what remains, who bears it, and why it is acceptable.

Decision and authority: release, restrict, delay or redesign; signed by the accountable officer, with the date of review.

T10. The National Position Canvas (Chapter 11)

When to use. To map a country's or region's position; complete the table and the five questions.

Use this canvas to map a country's position, or a region's, in the geography of AGI-era power. Rate each source of power as strong, moderate or weak, now and in 2035 on current trends. Use the table below. Then answer:

Worked example. Chapter 11: a middle-income country declined both frontier-state offers as presented, combining limited frontier access, a 24-month wind-down clause, sovereign inference and regional pooling at about a 20% premium.

Indispensable links: in which domain is the country, or could it become, essential to others?

Critical dependencies: which dependencies could be used as leverage against the country, and by whom?

Bargaining position: what do the frontier states need from the country?

Partners: with whom could the country pool resources?

Strategy mix: which of the five strategies (specialize, diffuse, pool, hedge, shape the rules) fit, and in what combination?

T11. The Escalation and Human-Control Review (Chapter 13)

When to use. For AI-enabled military, intelligence and cyber systems, and civilian systems whose actions another state could misinterpret.

Apply this review to any AI-enabled military, intelligence or cyber system, and to civilian systems whose actions could be misinterpreted by another state.

Worked example. Chapter 13: a cyber agency delegated defensive containment within national networks, reversible within 60 seconds; 37 of 1,140 actions were reversed in the first year, and four regional hospital disruptions were prevented.

Function: what the system detects, recommends or does.

Delegated authority: which actions it may take without human approval, and within what limits.

Reserved decisions: which decisions remain human (always including the use of force, actions affecting foreign systems and anything touching nuclear command and control).

Decision time: how long humans have to decide at each stage, and whether that is enough to understand the situation.

Speed-matched intervention: how quickly a human can stop or reverse the system's actions.

Misperception risk: how another state's systems might interpret these actions; what notification or communication channel exists.

Attribution: how the system distinguishes state actors, criminals and accidents, and with what confidence.

Automation bias controls: how operators are trained and tested to question the system.

Legal review: compliance with international humanitarian law and national law.

Accountability: the commander or official who answers for the system's actions, and the political authority that renews its delegation.

T12. The Verification Design Checklist (Chapter 14)

When to use. To assess any agreement, regulation or voluntary commitment on advanced AI.

Use this checklist to assess any proposed agreement, regulation or voluntary commitment on advanced AI.

Worked example. Chapter 14: a five-country pilot failed when a red team split a training run across four sites below per-site thresholds; the redesign defined thresholds per run and required chip attestation.

What is being limited or required? Compute, capability, deployment conditions, safety evaluations or something else.

Observable indicator: what can actually be observed or measured that would show compliance?

Verification layers: which of compute accounting, chip mechanisms, physical signatures, evaluations and inspections are used?

Evasion paths: distributed training, algorithmic efficiency, open release, offshore operation, subsidiaries. How is each addressed?

Threshold durability: how will thresholds be updated as algorithms improve?

Confidentiality: how is sensitive information protected from inspectors and other parties?

Misuse of the regime: could the verification mechanisms be used for espionage, sabotage or competitive advantage?

Coverage: which countries and companies are inside, and where could activity move?

Testing: has the regime been red-teamed? When, by whom, with what result?

Response to non-compliance: what happens when a violation is found, and who decides?

T13. The Concentration and Distribution Test (Chapter 15)

When to use. For major AI investments, acquisitions, markets, public programmes and national strategies.

Apply this test to a major AI investment, acquisition, market, public programme or national strategy.

Worked example. Chapter 15: an acquisition of an open-weight laboratory was approved with ten-year open-licence and equal-access conditions; an independent monitor later found a three-month head start for the acquirer's cloud, and a fine followed.

Concentration

Control: who controls the models, compute, data and distribution channels involved?

Switching: can users move to alternatives? At what cost?

Self-reinforcement: does success increase control (through data, scale or network effects)?

Alternatives: do open, public or competing alternatives exist and remain viable?

Distribution

Gains: who captures the productivity gains: owners, workers, consumers, the public?

Labour: does the system complement or substitute for workers? Which ones?

Geography: which regions and countries gain, and which lose?

Fiscal effect: how does it change the tax base?

Response

Conditions: what conditions (access, interoperability, portability, revenue sharing, public capacity) would improve the balance?

Monitoring and enforcement: who checks the conditions, and with what power?

T14. The Task Transition Map (Chapter 16)

When to use. For a role, team or occupation; complete the table and the five questions.

Complete for a role, team or occupation. Use the table below. Then record:

Worked example. Chapter 16: a retraining programme that taught entry-level programming placed fewer than a third of graduates; its successor trained for capabilities, not occupations.

Rising-value positions available in the field (from the seven in the AGI-Proof Career Framework).

Ladder risk: which beginner tasks are being automated, and how beginners will now build judgment.

Unassisted core: the competences that must be preserved without AI, and how they are practised.

Learning plan: capabilities to build, how, by when, with what evidence.

Signposts: capability developments that would require the map to be revised.

T15. The Distribution Mechanism Comparator (Chapter 17)

When to use. To design or evaluate strategies for sharing the gains of AI.

Use this comparator to design or evaluate a strategy for sharing the gains of AI. For each mechanism being considered, record:

Worked example. Chapter 17: a compute dividend of about €140 per adult missed 18% of eligible adults; automatic payment, a larger payment near the campuses and local public services raised coverage to 97%.

Who is reached: by design, and in practice; who is likely to be missed.

Who bears the costs: of AI's development and deployment (job loss, energy prices, land, data), and whether they share in benefits proportionately.

Scale: whether the benefit is large enough to matter to the people reached.

Speed: how quickly it can begin and grow.

Fiscal cost and funding source.

Durability: whether it would survive a change of government or an economic downturn.

Incentives: effects on investment, work and innovation.

Administration: the delivery channel, and whether it reaches people without digital access.

International dimension: effects on other countries.

Review: how and when its reach and effects will be measured.

A strategy should combine mechanisms so that everyone is reached, those who bear the costs are compensated, and the combination is durable.

T16. The Epistemic Security Audit (Chapter 18)

When to use. For a country, region, institution or platform.

Use this audit for a country, region, institution or platform.

Worked example. Chapter 18: a municipality offered AI companions to isolated older residents on five conditions, including protected funding for human visits; loneliness fell and visits were not reduced.

Sources: which sources do people rely on for news and public information? How many depend on AI systems?

Provenance: is provenance information attached to official, journalistic and campaign content? Is it visible to users?

Impersonation risk: which public officials, institutions and trusted voices could be impersonated, and how would people verify them?

Verification capacity: resources and AI tools available to journalists, electoral authorities and fact-checkers.

Influence operations: capacity to detect and respond to automated influence networks.

Citizens' capability: education in evaluating sources and recognizing manipulation, from school onward.

Vulnerable users: protections for children, older people and people in crisis, including from AI companions.

Concentration: whether a single AI system or company has become a gatekeeper of information.

Elections: rules on AI in campaigns, disclosure, and rapid response to synthetic media.

Trust measures: how public trust in information is measured over time.

T17. The National Strategy Canvas (Chapter 19)

When to use. To design or review a national, regional or sectoral AI strategy.

Use this canvas to design or review a national, regional or sectoral AI strategy.

Worked example. Chapter 19: of two similar countries, the one that invested in institutions and flexible compute outperformed the one that built an oversized national compute centre.

Position: the country's position from the National Position Canvas (Tool 10).

Ambition: the role the country seeks by 2035.

Must do ourselves: capabilities the country will hold domestically (including the sovereign minimum).

Will do with partners: capabilities obtained through partnership or pooling, and with whom.

Building blocks: status and plan for each of the eight (evaluation institute, procurement power, sovereign minimum, flexible compute, data and language, talent, public AI, authority rules).

Adoption priorities: three public services and three economic sectors where AI should be applied first, with measures.

Distribution: how the gains will be shared (Tool 15).

Protection: application of the 7-Step AGI-Resistant Architecture Protocol to state institutions.

International: institutions and coalitions to join or build.

Signposts and options: the signposts (Tool 3) that would trigger larger commitments, and the prepared options.

Review: who reviews the strategy, how often, and with what independence.

T18. The Board AGI Questions (Chapter 20)

When to use. Annually at board level.

Twelve questions every board should be able to answer, and revisit annually.

Worked example. Chapter 20: the hospital network's board answered Question 8 only after the February 2032 failure; the 30-Day Programme would have found the gap earlier.

Authority

Which AI systems can commit the company (to contracts, payments, communications, changes to records), and within what limits?

Are those limits technically enforced, and who tests them?

Where have we expanded AI autonomy in the past year, and on what evidence?

Dependency and continuity

Which critical functions depend on a single provider of models or compute?

If that provider failed or changed its terms tomorrow, what would stop, and for how long?

When did we last rehearse operating without it?

Safety and conduct

What were our most serious AI incidents this year, and what changed as a result?

Can every consequential system be stopped at the speed it operates, by someone always available?

Accountability and disclosure

For each consequential AI function, which executive answers for its outcomes?

How can a customer, employee or citizen harmed by our AI obtain a remedy?

What material AI risks and dependencies do we disclose, and are we confident the disclosure is complete?

Workforce and strategy

How are we forming the experienced people we will need in ten years, if AI now does the work through which they used to learn?

T19. The Institution Design Scorecard (Chapter 21)

When to use. To evaluate any proposed international or national institution for governing advanced AI.

Use this scorecard to evaluate any proposed international (or national) institution for governing advanced AI. Score each criterion from 0 (absent) to 3 (strong), with reasons.

Worked example. Chapter 21: the founding negotiations of an international agency accepted a transitional exemption for national security facilities and a mandatory benefit fund; 94 states signed.

Functions: which of the six functions (assessment, evaluation, verification, incident response, research and access, cooperation between rivals) does it perform?

Scope: does its mandate cover the systems that matter most, with thresholds that will remain meaningful?

Authority: what can it actually require, inspect or decide?

Accountability: to whom does it answer, and how can its decisions be challenged?

Participation: are the frontier states inside? Are developing countries represented with real influence?

Legitimacy: would those affected regard its decisions as rightful?

Verification: can compliance with its rules be confirmed?

Resistance to capture: are its own functions separated, audited and protected from misuse?

Adaptability: can it update its rules as capability changes?

Path: is there a realistic phased path from today's institutions to this one?

T20. The Accountability Chain Record (Chapter 22)

When to use. For every consequential AI function before deployment, and after every incident.

Complete for every consequential AI function, before deployment, and update after any incident.

Worked example. Chapter 22: after the February 2032 failure, the hospital network compensated a harmed patient within ten weeks without requiring proof of fault, then recovered costs from the vendor and the transport supplier.

Function and system.

Parties: model developer, data providers, configurer, deployer, operators, suppliers, interacting systems.

Accountable person: the named person or office that answers to those affected.

Duties: what each party is responsible for (testing, monitoring, disclosure, change notice, stop capability).

Contracts: clauses allocating responsibility among the parties (defects, change notice, indemnities, audit rights).

Rights of affected people: how they learn of AI involvement, obtain an explanation, request human review, appeal and obtain a remedy.

Redress: the scheme for compensating harm quickly, and its limits.

Incident reporting: what must be reported, to whom, within what time.

Evidence: logs and records preserved to establish what happened.

After an incident: what happened, who contributed, who compensated whom, what changed.

Part 3: Scenarios and Decisions

S1. The Scenario Workbook (Chapter 23)

Use this worksheet in a half-day or one-day workshop with a leadership team.

Worked example. Chapter 23: the four scenarios are Concentrated Acceleration, Turbulent Abundance, Platform Plateau and Gradual Diffusion, each with a better and a worse version.

Focal question: the decision or strategy to be tested (for example, "our ten-year national AI strategy").

Scenario immersion: for each of the four scenarios, describe what your organization or country looks like in 2030, 2035 and 2050, in the better and worse versions.

Stress test: how does your current strategy perform in each? Score it from 1 (fails) to 5 (thrives).

Vulnerabilities: in which scenario does it fail worst, and why?

Robust moves: which actions perform acceptably in all four?

Options and hedges: which prepared options would let you adapt if a particular scenario arrived?

Signposts: which developments would tell you which scenario is arriving? Add them to your Signpost Register (Tool 3).

Wildcards: how would each wildcard affect you, and what would you do in the first week?

Decisions: what will you change now?

S2. The Compassionate Decision Record (Chapter 24)

Complete before a significant decision involving AI, and again at review.

Worked example. Chapter 24: on the Heatwave Night, floors for essential services, life-dependent households and reachable cooling ruled out both proposed options; a prepared hybrid (1,250 MW of compute moved, 150 MW of flexible load, 50 MW of one-hour residential rotation) met every floor, the preferred action changed in each rehearsed failure, and the review named the residual harm and changed the rules.

Decision and objective: what is being decided, and the legitimate objective.

Options presented: including those proposed by AI systems; check that each is physically feasible, with units stated (for example MW and MWh).

Affected groups: everyone affected, including those absent and future generations; specifically those least able to protect themselves.

Floors: protections against severe, irreversible harm that no option may breach, written as constraints.

Order of comparison: the stated order in which options meeting every floor will be compared (severe-harm exposure first; then other serious burdens; distribution; reversibility and recovery; cost), each in its own units.

Feasible alternatives: options beyond those presented, and the preparation they require.

Comparison: each option against each floor and criterion; any departure from the stated order only through the exception process, with grounds, majority, evidence and reviewer recorded.

Sensitivity: which assumptions, if wrong, would change the preferred option, and the prepared fallback for each.

Infeasibility: if no option meets every floor, who declares it, under what process, and the least-harm fallback.

Participation: who set the floors, who was consulted, who decided.

Decision, residual burden and remedy: who is still harmed, how, and the compensation, support and correction owed.

Compassion Quotient review: how well the decision attended to suffering, avoided severe harm and cared for the vulnerable, as a diagnostic, never overriding a floor.

Review date and rule changes.